

Seminar 0: Review of Complex Numbers

Basic Practice

Purpose. These exercises review only the basic concepts and operations with complex numbers that students are expected to know from previous courses.

1. The imaginary unit and quadratic equations

Recall that $i^2 = -1$. Solve the following equations over the complex numbers.

- (a) $x^2 + 9 = 0$
- (b) $x^2 - 2x + 5 = 0$
- (c) $5x^2 - 2x + 1 = 0$
- (d) $x^3 + 2x^2 + x = 0$

2. Real and imaginary parts

For each complex number, identify its real and imaginary parts.

- (a) $z_1 = 1 + i$
- (b) $z_2 = 3 - 2i$
- (c) $z_3 = -4i$
- (d) $z_4 = 7$
- (e) $z_5 = -2 + 5i$

Then write each number in the form $z = x + iy$, with $x, y \in \mathbb{R}$.

3. Addition, subtraction, and multiplication

Let $z_1 = 1 + i$, $z_2 = 3 - 2i$.

Compute:

- (a) $z_1 + z_2$
- (b) $z_1 - z_2$
- (c) $z_2 - z_1$
- (d) $z_1 z_2$
- (e) z_1^2
- (f) z_2^2

Express every result in Cartesian form.

4. Division of complex numbers

Compute and express the results in Cartesian form.

- (a) $\frac{1+i}{3-2i}$
- (b) $\frac{2-i}{1+i}$
- (c) $\frac{1}{2+3i}$

(d) $\frac{4i}{1-i}$

(e) Then solve $(3 - 2i)z = 1 + i$.

5. Powers of the imaginary unit

Compute:

(a) i^0, i^1, i^2, i^3, i^4

(b) i^{17}

(c) i^{38}

(d) i^{2001}

(e) $(-i)^{23}$

(f) State the general rule for evaluating i^n for any integer $n \geq 0$.

6. Complex conjugation

For each number, compute its complex conjugate.

(a) $1 + i$

(b) $3 - 2i$

(c) $-4i$

(d) 5

(e) $\sqrt{3} + 2i$

Let $z = x + iy$.

(f) Show that $z\bar{z} = x^2 + y^2$.

(g) Also verify that $\overline{z_1 + z_2} = \bar{z}_1 + \bar{z}_2$

(h) And $\overline{z_1 z_2} = \bar{z}_1 \bar{z}_2$.

7. Recovering real and imaginary parts

Let $z = x + iy$.

(a) Show that $\operatorname{Re}(z) = \frac{z + \bar{z}}{2}$

(b) And $\operatorname{Im}(z) = \frac{z - \bar{z}}{2i}$.

(c) Use these formulas to find the real and imaginary parts of $z = 7 - 5i$.

8. Modulus and distance

Compute the modulus of each complex number.

(a) $1 + i$

(b) $3 - 4i$

(c) $-5i$

(d) $-2 + 2i$

(e) $\sqrt{3} + i$

Then, for $z = x + iy$,

express the following quantities in terms of x and y :

- (f) $|z + 7i|$
 (g) $|z - 2|$
 (h) $|z - (5 + i)|$
 (i) Interpret each expression as a distance in the complex plane.

9. Cartesian and polar forms

For each complex number, compute its modulus r and an argument θ , and write it in polar form $z = re^{i\theta}$.

- (a) $1 + i$
 (b) $3 - 2i$
 (c) $1 + \sqrt{3}i$
 (d) $-1 + i$
 (e) $-2 - 2i$
 (f) $\frac{1+i}{1-i}$

- (g) Represent each number graphically in the complex plane.

10. Euler's formula

- (a) Using the Taylor series of the exponential, cosine, and sine functions, derive Euler's formula $e^{i\theta} = \cos\theta + i\sin\theta$.
 (b) Then use it to show that $\cos\theta = \frac{e^{i\theta} + e^{-i\theta}}{2}$
 (c) And $\sin\theta = \frac{e^{i\theta} - e^{-i\theta}}{2i}$.

11. Multiplication in polar form

Let $z_1 = r_1 e^{i\theta_1}$, $z_2 = r_2 e^{i\theta_2}$.

- (a) Show that $z_1 z_2 = r_1 r_2 e^{i(\theta_1 + \theta_2)}$.
 (b) Conclude that $|z_1 z_2| = |z_1| |z_2|$
 (c) and that the argument of a product is the sum of the arguments, modulo 2π .
 (d) Verify the result for $z_1 = 1 + i$, $z_2 = \sqrt{3} + i$.

12. De Moivre's formula

- (a) Use Euler's formula to prove that $(\cos\theta + i\sin\theta)^n = \cos(n\theta) + i\sin(n\theta)$

for every positive integer n .

Use De Moivre's formula to compute:

- (b) $(1 + i)^8$
 (c) $(\sqrt{3} + i)^6$
 (d) $(1 - i)^{10}$
 (e) Give each result in both polar and Cartesian form.

13. Roots of complex numbers

Find all solutions of the following equations and represent them in the complex plane.

- (a) $z^3 = 1$
 (b) $z^4 = 1$

- (c) $z^4 = -1$
- (d) $z^6 = 64$
- (e) For each problem, state the modulus and arguments of all roots.

14. Geometric interpretation

For each transformation, describe geometrically what happens to a point z in the complex plane.

- (a) $z \mapsto iz$
- (b) $z \mapsto -z$
- (c) $z \mapsto \bar{z}$
- (d) $z \mapsto 2z$
- (e) $z \mapsto e^{i\pi/4}z$
- (f) $z \mapsto 2e^{i\pi/3}z$
- (g) Apply each transformation to $z = 1 + i$.

15. Basic identities and inequalities

Let $z, w \in \mathbb{C}$. Prove or verify the following statements.

- (a) $|z|^2 = z\bar{z}$
- (b) $|\bar{z}| = |z|$
- (c) $|zw| = |z| |w|$
- (d) $\left| \frac{z}{w} \right| = \frac{|z|}{|w|}$ for $w \neq 0$
- (e) $|z + w| \leq |z| + |w|$
- (f) $|z - w|$ is the distance between the points represented by z and w
- (g) Illustrate the triangle inequality with a drawing in the complex plane.